

RANGA FARMS

Goat Farm Management System



Product Specification & Feature Documentation

A comprehensive digital solution for livestock, operations, and financial tracking.

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1. Product Purpose & Overview

Ranga Farms - Goat Farm Management System is a comprehensive, full-stack web application designed specifically to streamline, digitize, and optimize all aspects of managing professional goat farm operations. Built using a robust Flask backend, PostgreSQL database, and responsive Bootstrap interface, the system acts as a single point of truth for farm owners, managers, veterinarians, and workers.

By replacing paper logbooks and fragmented spreadsheets, Ranga Farms provides automated tracking of livestock lifecycles, real-time financial health monitoring, granular HR and work logs, veterinary medicine inventory control, feed planning, and rich reporting dashboards.

1.1 Technology Stack

- Backend: Flask 3.0.0 (Python 3)
- Database: PostgreSQL (with dynamic schema migration and connection pooling)
- Frontend: HTML5, CSS3, JavaScript, Jinja2 template engine, Bootstrap UI
- Authentication: Secure password hashing (Werkzeug) & OTP-based verification (pyotp)

2. Core Features & Functional Modules

2.1 Livestock Management

- Goat Master Registry: Comprehensive records of each goat, including Tag IDs, breed, age, status, gender, and parents.
- Kid Tracking: Monitors newborn goats, births, and parental lineage.
- Breed Management: Cataloging different goat breeds.
- Mortality Records: Accurate logging of deaths with causes for analysis.

2.2 Financial Management

- Profit & Loss (P&L): Real-time analysis of income and expenses.
- Sales & Purchases: Detailed logging of buying/selling goats, feed, medicine, vaccines, and equipment.
- Invoicing: Generating professional bills and invoice documents.
- Recurring Expenses: Automating monthly utilities, rent, or scheduled payments.

2.3 Employee Directory & HR

- Directory: Detailed worker roles, contacts, and metadata.
- Attendance & Leaves: Daily clock-in/out tracking and vacation requests.
- Salary Management: Automating monthly/daily wage calculations and payslip exports.
- Work Logs: Assigning and tracking daily tasks across the farm.

2.4 Health & Veterinary Care

- Medical Logs: Individual treatment history, symptoms, and outcomes.
- Doctor Management: Directory of visiting vets and schedules.
- Vaccine Schedule: Tracking upcoming and historical vaccinations.
- Medicine Inventory: Tracking stock levels and alerts for low inventory.

3. System Workflow & Recent Enhancements

The system is designed with a standard daily operational workflow:

1. **Authenticate:** Users sign in using credentials and optional OTP secondary authentication.
2. **Actionable Dashboard:** Displays immediate metrics: total goats, monthly profit/loss, active tasks, and recent alerts.
3. **Live Search & Operations:** Operations teams search goats or log transactions directly from the top interface.

Recently, two major usability enhancements were added to improve daily productivity:

Weight Notification Alert (25 kg or More)

To identify market-ready goats, the system automatically checks goat weights upon login. If any active goat has reached 25 kg or more, a sleek, modern popup modal alert displays on the user's first login of the day. The alert showcases the goat's Tag ID, weight, and color, and will not reappear for that user until the next day once dismissed.

Enhanced Tag ID Search (Last 4 Digits Support)

Finding a specific goat is fast and intuitive. Users can search the directory or dashboard using only the last 4 digits of the Tag ID (e.g., searching '0101' will match full Tag ID 'UII1710000101'). The search intelligently distinguishes between 4-digit numeric patterns, breed names, and full Tag IDs.